

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping

Course Code:	CSA0008	Course Title:	Data Structures
CO Assessed:	CO1 – Iteration (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
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Section / Batch:	CSE	Date:	15.07.2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

This test contains 2 concept mapping, Total: 100 Marks.

Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification, Linkages, Organization, Integration of Literature, and Clarity

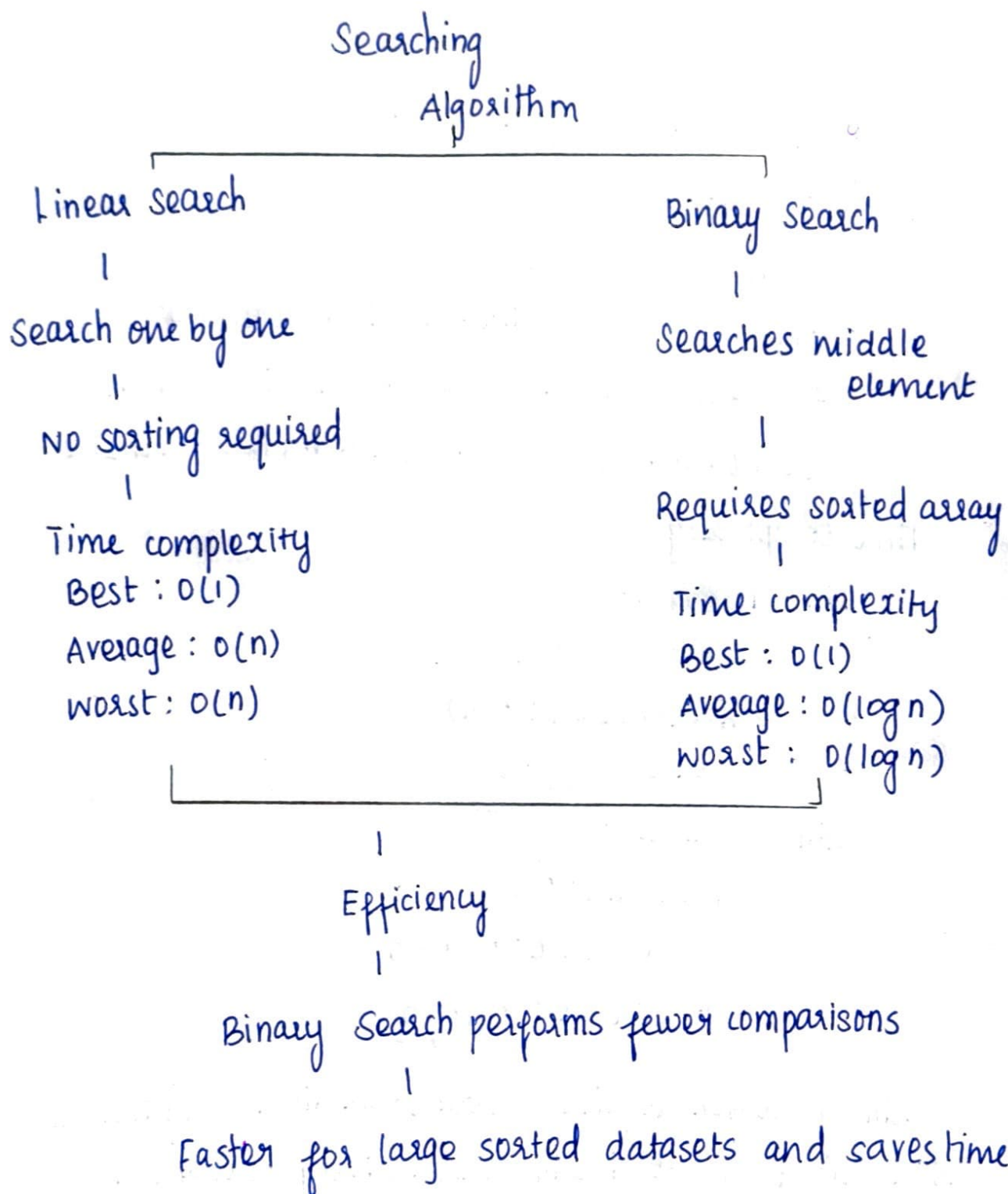
No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

Q11. Searching Algorithm - Linear Search $\rightarrow$ Binary Search - Time Complexity $\rightarrow$ Efficiency.

Concept Map:



A searching algorithm is used to locate an element in a collection of data.

Linear Search

checks each element one by one

works on both sorted and unsorted arrays

Easy to implement

Requires more comparisons for large datasets

Binary search:

works only on sorted arrays

compares the key with the middle element

Eliminates half of the search space in every step

Much faster than linear search for large datasets.

Time Complexity:

Linear search:

Best case: $O(1)$

Average case: $O(n)$

Worst case: $O(n)$

Binary search:

Best case: $O(1)$

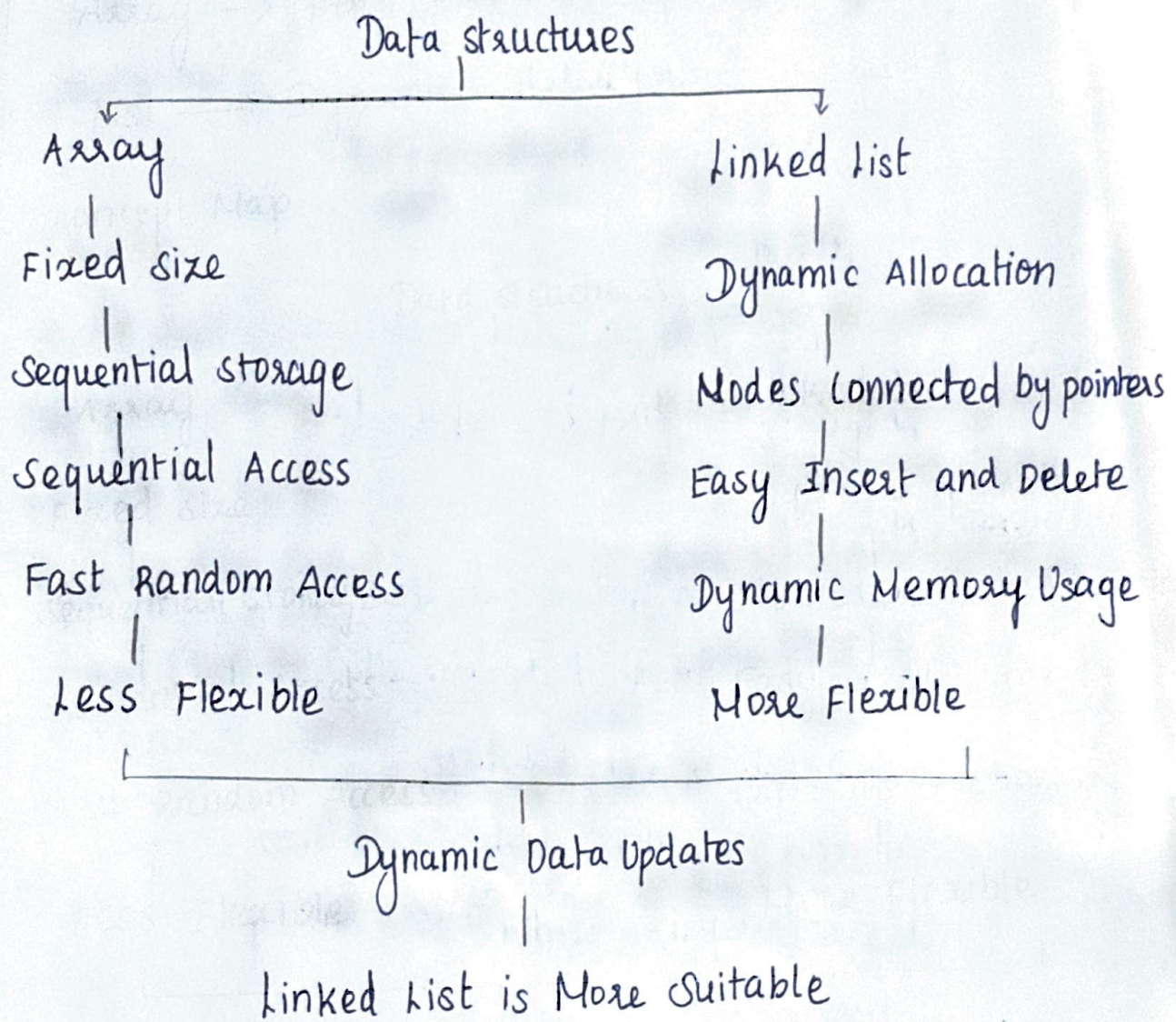
Average case: $O(\log n)$

Worst case: $O(\log n)$

Binary Search requires fewer comparisons and less execution time. As the input size increases, Binary Search performs much better than linear search.

Array $\rightarrow$ Fixed size $\rightarrow$ Sequential Access $\rightarrow$ Linked List $\rightarrow$ Dynamic Allocation $\rightarrow$ Flexibility.

concept Map:



An array stores elements in contiguous memory locations.

Array:

- Fixed size

- Fast access using index

- Sequential storage

- Easy to traverse

- Difficult to insert or delete elements because shifting is required.

Linked List

- Stores data in nodes connected using pointers

- Memory is allocated dynamically

- Easy insertion and deletion

- can grow or shrink during execution

- Does not require contiguous memory

Sequential Access:

Array allows sequential traversal and direct indexing, making access faster.

Dynamic Allocation:

Linked lists allocate memory only when needed, reducing memory wastage

Flexibility: Linked lists are more flexible because